

Exponentials & Polynomials



Laws of exponents

- 1 Simplify $x^5 \cdot x^3$.
 - 2 Simplify $\frac{y^9}{y^4}$.
 - 3 Simplify $(2x^3)^4$.
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Negative and fractional exponents

- 4 Simplify x^{-3} , writing your answer without a negative exponent.
 - 5 Simplify $16^{3/4}$.
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Adding, subtracting, and multiplying polynomials

- 6 Simplify $(3x^2 + 2x - 5) + (x^2 - 4x + 7)$.
 - 7 Expand $(x + 3)(x - 5)$.
 - 8 Expand $(2x - 1)^2$.
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Factoring polynomials

- 9 Factor $x^2 - 16$ completely.
 - 10 Factor $2x^2 + 8x$ completely.
 - 11 Factor $x^3 - 4x$ completely.
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Polynomial division and the remainder theorem

- 12 Find the remainder when $f(x) = x^3 - 2x^2 + x - 5$ is divided by $(x - 2)$, using the remainder theorem.
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Exponential growth vs. polynomial growth

- 13 Two functions are given: $f(x) = x^3$ and $g(x) = 2^x$. Explain which function eventually grows faster as x increases, and estimate the crossover point using the values $f(10) = 1000$, $g(10) = 1024$, $f(8) = 512$, $g(8) = 256$.
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Modeling with exponential functions

- 14 An investment of 5000 dollars grows according to $A(t) = 5000(1.06)^t$, where t is the number of years. Find the value of the investment after 5 years, to the nearest dollar, and state the annual growth rate as a percentage.
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Applying the zero product property to polynomial equations

- 15 Solve $x^3 - 5x^2 + 6x = 0$ by factoring completely and applying the zero product property.
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Multiplying binomials and trinomials

- 16 Expand $(x + 2)(x^2 - 3x + 4)$.
- 17 Expand $(2x - 3)(x + 5)$.
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Comparing exponential models

- 18 Two savings accounts grow according to $A(t) = 1000(1.03)^t$ and $B(t) = 1000(1.05)^t$. Explain which account grows faster, and find the difference in value after 10 years, to the nearest dollar.
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Simplifying expressions with multiple exponent rules

- 19 Simplify $\frac{(x^3y^2)^2}{x^4y}$.
- 20 Simplify $\left(\frac{2x^2}{y}\right)^3$.
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Half-life and decay applications

- 21** A radioactive substance has a half-life of 8 years, starting with 200 grams. Write a model for the remaining amount after t years, and find the amount remaining after 24 years.